

Algebraic Composition Limits in FFGFT

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Abstract

In FFGFT, particle masses are derived from an algebraic torus structure: $m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v$. These terms implicitly carry the geometry of their origin. When inserted into formulas from a different algebraic context — such as the Koide relation or Standard Model scattering amplitudes — they generate structure-alien cross-terms that belong to neither the FFGFT geometry nor the target formula. This error is not a numerical problem but a category error: it violates the composition limits between different algebraic structure spaces. This document analyses the error systematically, shows its manifestations in FFGFT, and states the correct rule for handling theory-internal terms in external formulas.

.1 The Problem: Terms versus Results

Two levels in FFGFT

FFGFT operates on two clearly separated levels:

Structure level (terms): Symbolic expressions such as $m_\tau = r_\tau \cdot \xi^{2/3} \cdot v$ describe the *geometric origin* of the mass. They encode the torus topology (exponent $p = 2/3$), the recursion depth (prefactor $r_\tau = 25/9$) and the scale (vacuum expectation value v).

Result level (numbers): The numerical value $m_\tau \approx 1785 \text{ MeV}$ is a number without structural memory. It can be inserted like a measured value into any formula.

The rule: Results (numbers) may be inserted anywhere. Terms (algebraic expressions) may only be used within the algebraic space from which they originate.

The Koide example

The Koide relation reads:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (1)$$

This is experimentally confirmed to 0.001 % (PDG values).

Permitted use

One evaluates the FFGFT terms numerically:

$$\begin{aligned} m_e &\approx 0.5054 \text{ MeV}, \\ m_\mu &\approx 105.05 \text{ MeV}, \\ m_\tau &\approx 1785.0 \text{ MeV}, \end{aligned}$$

(the FFGFT values $r_i \xi^{p_i} \nu$ with $\nu = 246.22 \text{ GeV}$) and inserts these numbers into (1). One obtains $Q_{\text{FFGFT}} \approx 0.6677 \neq 2/3$, a deviation of 0.16 %. This is correct and honest: the FFGFT masses are approximations ($\sim 1\%$ accuracy), and this imprecision propagates into Q .

Not permitted

One inserts the FFGFT terms *symbolically* into (1):

$$Q \stackrel{?}{=} \frac{r_e \xi^{3/2} + r_\mu \xi + r_\tau \xi^{2/3}}{(\sqrt{r_e \xi^{3/2}} + \sqrt{r_\mu \xi} + \sqrt{r_\tau \xi^{2/3}})^2} \stackrel{?}{=} \frac{2}{3} \quad (2)$$

and attempts to verify this equation algebraically.

This fails. The reason is analysed in the next section.

.2 Cross-term Analysis

Where do the cross-terms come from?

The denominator of Q contains $(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$. Inserting the FFGFT terms:

$$\begin{aligned} \sqrt{m_e} &\propto \xi^{3/4}, \\ \sqrt{m_\mu} &\propto \xi^{1/2}, \\ \sqrt{m_\tau} &\propto \xi^{1/3}, \end{aligned}$$

squaring the denominator yields three *diagonal terms*:

$$m_e + m_\mu + m_\tau \propto \xi^{3/2} + \xi + \xi^{2/3} \quad (3)$$

(the same powers as in the numerator) — and three *cross-terms*:

$$2\sqrt{m_e m_\mu} \propto \xi^{3/4+1/2} = \xi^{5/4}, \quad (4)$$

$$2\sqrt{m_e m_\tau} \propto \xi^{3/4+1/3} = \xi^{13/12}, \quad (5)$$

$$2\sqrt{m_\mu m_\tau} \propto \xi^{1/2+1/3} = \xi^{5/6}. \quad (6)$$

Why these cross-terms are structure-alien

The exponents in the FFGFT structure space form an explicit list:

$$p \in \left\{ \frac{2}{3}, 1, \frac{3}{2}, \frac{1}{3}, \frac{1}{2}, \frac{3}{4} \right\}.$$

The cross-term exponents $5/4$, $13/12$, $5/6$ do *not* belong to this list. They arise only through the square-root operation of the Koide formula and have no counterpart in the torus geometry. This separation holds only against the explicit list, not via divisibility: $3/2$, $1/2$ and $3/4$ are not multiples of $1/3$ (already $p_e = 3/2$ is not in $\frac{1}{3}\mathbb{Z}$); the least common denominator of the list is 12, and $\frac{1}{12}\mathbb{Z}$ also contains $5/4$, $13/12$ and $5/6$.

Structure-alien terms

Cross-terms from the Koide operation lie outside the FFGFT structure space. They are neither geometrically interpretable nor physically meaningful. They are an artefact of formula composition.

Condition for $Q = 2/3$ exactly

For $Q = 2/3$ to hold algebraically exactly, one would need:

$$m_e + m_\mu + m_\tau = 4 \left(\sqrt{m_e m_\mu} + \sqrt{m_e m_\tau} + \sqrt{m_\mu m_\tau} \right). \quad (7)$$

This equation links the diagonal terms to the cross-terms. It is not satisfied for $\xi = 4/30000$. The ξ -value for which it holds exactly is $\xi_{\text{Koide}} \approx 1.372 \times 10^{-4}$ — 2.9 % larger than the geometric value $\xi_{\text{par}} = 4/30000$.

This shows: the Koide relation enforces an algebraic constraint that is independent of the FFGFT torus geometry.

.3 The General Principle: Composition Limits

Algebraic structure spaces

Every physical theory defines an *algebraic structure space*: the set of expressions that are well-formed within the theory, and the operations that are permitted within this space.

FFGFT structure space: Expressions of the form $r \cdot \xi^p \cdot v$ with p from the explicit exponent list of the previous section (not from $\frac{1}{3}\mathbb{Z}$, since already $p_e = 3/2$ is not in it). Permitted operations: multiplication, division and exponentiation, as long as the exponents remain in this list.

Koide space: Formula (1) with the operation $\sqrt{\cdot}$. This operation leaves the FFGFT structure space: $\sqrt{\xi^{2/3}} = \xi^{1/3}$ is still within it, but $\xi^{1/3} \cdot \xi^{1/2} = \xi^{5/6}$ is no longer in the exponent list.

Composition limit

Two algebraic structure spaces $\mathcal{A}$ and $\mathcal{B}$ are *compositionally compatible* if all operations in $\mathcal{B}$ leave the space $\mathcal{A}$ invariant. Otherwise a composition limit exists: terms from $\mathcal{A}$ must not be symbolically inserted into formulas from $\mathcal{B}$.

Further manifestations in FFGFT

The Koide example is not unique. The same problem arises with:

Quantum numbers in Standard Model formulas

FFGFT quantum numbers (torus winding numbers, Z_3 charges) have a specific algebraic origin. They must not be inserted directly into renormalisation equations or SM scattering amplitudes, as these formulas presuppose different algebraic conventions.

Unit systems as a special case

The most familiar example of composition limits is unit systems. An expression in natural units ($\hbar = c = 1$) must not be inserted without conversion into an SI formula. FFGFT terms carry an analogous implicit “unit”: the torus geometry.

Scale hierarchies

The FFGFT terms $\xi^{p_i} \cdot v$ are calibrated for the particle sector. They must not be inserted without scale correction into cosmological formulas, where different levels of the ξ -hierarchy apply.

The correct procedure

Composition rule

Step 1: Evaluate the FFGFT term numerically (result as a number).

Step 2: Insert that number like a measured value into the target formula.

Step 3: Honestly declare the result as an approximation when the term itself is only an approximation.

This is structurally identical to error propagation in experimental physics: before inserting a value into a new formula, one separates number and uncertainty. The uncertainty of the FFGFT term ($\sim 1\%$) propagates — and produces a different deviation in nonlinear formulas such as Koide than in linear ones.

.4 Consequences for FFGFT Documentation

What the Koide deviation means

The deviation $Q_{\text{FFGFT}} \approx 0.6677 \neq 2/3$ is not an error of the theory and not a contradiction of the experimental confirmation of the Koide relation. It is the correct propagation of the FFGFT mass accuracy ($\sim 1\%$) through a nonlinear formula with a cross-term problem.

The experimental Koide relation $Q = 2/3$ confirms the FFGFT exponent structure $\Delta p = 1/3$ by a different route: it shows that the *true* masses satisfy condition (7) — and that the FFGFT terms approach this condition to 0.16 % without enforcing it.

Linguistic consequence

In external communications, a distinction must be made:

Statement	Status	Reason
"FFGFT explains the Koide formula through the exponent step $1/3$ "	Correct	Structural statement about origin
"FFGFT calculates $Q = 2/3$ exactly"	Incorrect	Composition limit violation
"FFGFT masses agree with the Koide relation to 0.16 %"	Correct	Honest numerical statement
"The Koide relation is only valid with PDG values"	Correct	Context of the original formula

Connection to Doc. 193

Document 070 (*On the Mathematical Structure of T0 Theory: Why Products of Different r_i Are Structurally Meaningless*) addresses the same fundamental principle at the level of mass formulas.

Doc. 193 proves the **non-closure theorem**: the set of admissible pairs (r_i, p_i) in FFGFT is not closed under multiplication:

$$\forall i \neq j : (r_i \cdot r_j, p_i + p_j) \notin \mathcal{P}$$

This means: $r_e \cdot r_\mu = 4/3 \cdot 16/5 = 64/15$ belongs to no torus mode and no particle in FFGFT. The r_i are not numbers in a field but labels of a discrete geometric space — their composition is not ordinary field multiplication.

Doc. 193 shows: *Mass formulas must not be inserted symbolically into other formulas, because this generates cross-products $r_i \cdot r_j$ that lie outside $\mathcal{P}$.*

Doc. 192 (this document) generalises: *Terms from one algebraic structure space must not be symbolically inserted into formulas of a foreign structure space* — this applies to the Koide relation, to quantum numbers in SM formulas, and to all scale hierarchies in FFGFT.

Both documents describe the same composition limit: Doc. 193 at the level of mass formula terms (non-closure theorem), Doc. 192 at the level of formula composition between different theories.

Summary

The composition limit between the FFGFT structure space and the Koide formula explains the observed 0.16 % deviation in Q . It is not a theory error but an indication of a categorical distinction: structural terms (with implicit geometry) and numerical results (numbers without structural memory) must not be used synonymously in foreign algebraic contexts.

The principle is general: it applies to quantum numbers in SM formulas, to unit system transitions, and to scale hierarchies in FFGFT. The correct procedure is always: evaluate first, then insert.